GGG-P-480E
March 11, 1974
SUPERSEDING
Int. Fed. Spec. GGG-P-00480D(GSA-FSS)
May 26, 1972 and
Fed. Spec. GGG-P-480A
February 8, 1960

FEDERAL SPECIFICATION

PLIERS, RETAINING RING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pliers used for inserting and removing internal and external retaining rings.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Pliers shall be of the following types, classes, and styles as specified (see 6.1):

Type I - Retaining ring, internal.
- Class 1 - With adjustable stop and spring.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.
- Class 2 - Without adjustable stop and spring.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.
- Class 3 - With ratchet, without adjustable stop.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.
- Class 4 - With ratchet and adjustable stop.
  - Style D - 15° bent tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.

Type II - Retaining ring, external.
- Class 1 - Fixed jaw stop and spring.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.
- Class 2 - With adjustable stop and spring.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.
- Class 3 - Without adjustable stop and spring.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.
- Class 4 - With ratchet, without adjustable stop.
  - Style A - Straight tip.
  - Style B - 90° bent tip.
  - Style C - 45° bent tip.

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Class 5 - With ratchet and adjustable stop.
Style D - 15° bent tip.
Style B - 90° bent tip.
Style C - 45° bent tip.
Class 6 - Grip ring plier, with adjustable stop and spring.
Style E - 20° bent tip.
Style B - 90° bent tip.
Style C - 45° bent tip.

Type III - Retaining ring, external (automotive).
Class 1 - Straight jaw.
Class 2 - Brake key.

Type VII - Plier sets with interchangeable points.
Class 1 - With spring.
Class 2 - With ratchet.

Type VIII - Retaining ring, internal and external (universal).
Class 1 - Without adjustable stop or spring.
Style A - Straight tip.
Style B - 90° bent tip.
Style C - 45° bent tip.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

FPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:

Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals).
MIL-R-21248 - Rings, Retaining (Tapered and Reduced Section Type).

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive for dimensional data only and are not intended to preclude pliers otherwise in accordance with this specification. The overall length dimension for pliers with integral tips is intended to apply to those having straight tips. Bent tip pliers by their design may fall outside of the tolerance range. Dimensions shown on illustrations are in inches.

3.2 Material. Unless otherwise specified hereinafter, the pliers shall be of steel, of such composition and quality that when properly heat treated where necessary, will produce tools conforming to the requirements of this specification.

3.3 Handles. Handles shall be smooth, of a rounded channel-section type, or shall be covered with a cushioning material to provide a comfortable handgrip.

3.4 Joints. The pliers shall have riveted joints and shall pivot freely on the hinge pin with no side play in the joint. At the option of the manufacturer, washers may be added to the joint for additional bearing surface.

3.5 Jaws. The jaws shall operate through the required plier movement without binding and shall be either integral with, riveted, or welded to the handles. The tips of the jaws shall be either integral with, riveted, welded, or joined to the jaws by use of screws and nuts. The tip ends shall be round in cross section.

3.6 Tolerance. A tolerance of plus or minus 5° will be permitted on the bent tips.

3.7 Ring sizes accommodated. Rings, which these pliers are intended to operate, are covered by MIL-R-21248. The pliers shall be capable of installing and removing retaining rings in the range of sizes indicated for the specific type and size of plier.

3.8 Finish.

3.8.1 Surfaces. All surfaces shall be free from pits, nodules, burrs, cracks, and other defects. Surfaces not usually ground or buffed shall have a surface conforming to good manufacturing practices.

3.8.2 Coating. The pliers shall have one, or a combination, of the following coatings or platings, at the option of the contractor.

3.8.2.1 Oxide coating. The coating shall consist of a heavy, chemically-produced oxide.

3.8.2.2 Phosphate coating. The coating shall consist of a heavy, chemically-produced phosphate.

3.8.2.3 Chromium plate. The finish shall be electrodeposited metals of nickel, followed by chromium, the minimum thickness being 0.0002 inch for nickel and 0.00001 inch for chromium.

3.8.2.4 Zinc. The coating shall be mechanically or electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.9 Marking. The pliers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined, and with the plier size, except for type VII. Due to the interchangeable point sizes, type VII pliers do not require size marking. Pliers shall be marked before coating (see 3.8.2) so that the tools will have a complete, unbroken coating.

3.10 Type I, pliers, retaining ring, internal. Type I pliers shall be of steel, except as noted. They shall be suitable for compressing internal retaining rings similar to those shown in figure 1, and shall satisfactorily contract test rings when tested as specified in 4.5.1.

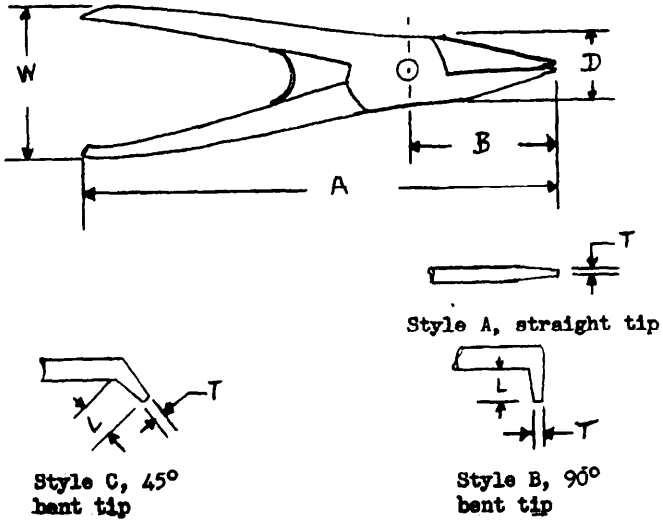
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Basic Type Inverted Type

FIGURE 1. Internal retaining rings.

3.10.1 Type I, class 1, with adjustable stop and spring. Type I, class 1 pliers shall consist of two jaw-handle sections joined by a rivet. An adjustable stop and spring shall be provided. The pliers shall be furnished in styles A, B, or C, as specified in 3.10.1.3 through 3.10.1.5, and 6.1 Plier sizes, as specified (see 6.1), shall conform to the dimensions in figure 2 and table I.



Style A, straight tip

Style C, 45° bent tip

Style B, 90° bent tip

FIGURE 2. Type I, class 1, styles A, B, and C pliers, retaining ring, internal, with adjustable stop and spring, straight and bent tip.

TABLE I. Type I, classes 1 and 2, styles A, B, and C pliers, retaining ring, internal, with and without adjustable stop and spring, straight and bent tip

Plier size		Housing diameter of rings		A ±1/2	B ±1/4	D ±3/16	T +.002 -.005	W ±1/2	L (Styles B and C)	Test ring size
Class 1	Class 2	Basic, bowed or beveled type	Inverted type							
		Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inch	Inches
1-1-1	1-2-1	0.250 thru .312	----	5-7/8	1-7/8	7/8	0.025	1-5/8	9/32 ±1/16	0.312
1-1-2	1-2-2	0.375 thru .562	0.750 thru 1.000	5-7/8	1-7/8	7/8	0.038	1-5/8	9/32 ±1/16	1/ .562 1.000
1-1-3	1-2-3	0.625 thru 1.023	1.062 thru 1.375	5-7/8	1-7/8	7/8	0.047	1-5/8	9/32 ±1/16	1/ 1.023 1.375
1-1-4	1-2-4	1.062 thru 1.750	1.438 thru 2.000	6-7/16	2-1/8	7/8	0.070	2	11/32 ±1/16	1/ 1.750 2.000
1-1-5	1-2-5	1.812 thru 3.000	2.062 thru 3.000	8-7/8	2-3/4	1-1/8	0.090	2-5/8	7/16 ±1/8	1/ 3.000 3.000

1/ Inverted type ring.

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3.10.1.1 Spring. The spring shall be of music wire, or hard-drawn spring-tempered steel wire in accordance with good commercial practice. The spring shall be capable of returning the handles to an open position after they have been compressed.

3.10.1.2 Adjustable stop. An adjustable stop shall be provided for adjusting and limiting the jaw travel to any desired setting within the prescribed range of plier use. The stop shall not extend beyond the outer edges of the jaw if it is located between the rivet and the tip end of the jaw.

3.10.1.3 Style A, straight tip. The tip end of the jaws of style A pliers shall be straight.

3.10.1.4 Style B, 90° bent tip. The tip end of the jaws of style B pliers shall be bent at a 90° angle to the longitudinal axis of the jaws.

3.10.1.5 Style C, 45° bent tip. The tip end of the jaws of style C pliers shall be bent at a 45° angle to the longitudinal axis of the jaws.

3.10.2 Type I, class 2 pliers shall consist of two jaw-handle sections joined by a rivet and shall be without adjustable stop and spring. They shall be furnished in styles A, B, or C, as specified in 3.10.1.3 through 3.10.1.5, and 6.1. Plier sizes, as specified (see 6.1), shall conform to the dimensions in figure 3 and table I.

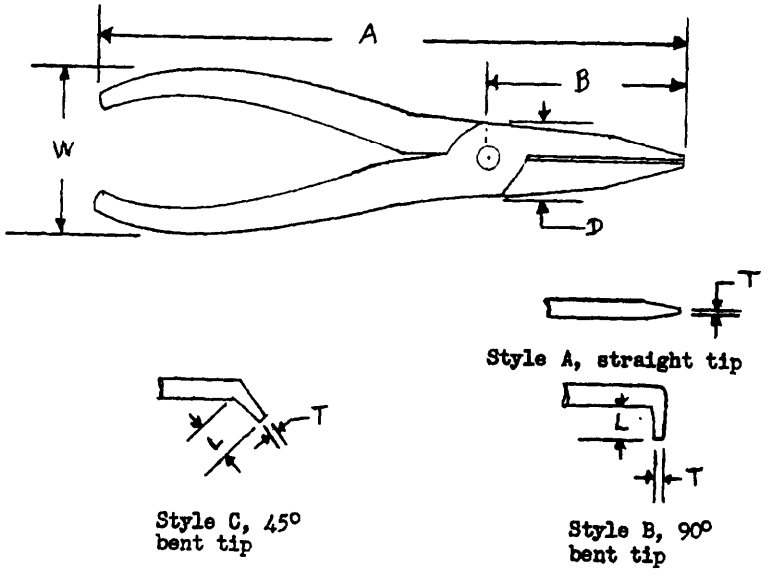


FIGURE 3. Type I, class 2, styles A, B, and C pliers, retaining ring, internal, without adjustable stop and spring, straight tip and bent tip.

3.10.3 Type I, class 3, with ratchet, without adjustable stop. Type I, class 3 pliers shall consist of two jaw-handle sections joined by a rivet, and shall be equipped with a ratcheting device. It shall be made of steel, but may be fitted with a handle of other suitable material. The ratcheting device shall act as a safety device to retain a ring in the stressed condition until released by actuating the handle. The ratchet shall be designed to permit release while the hand maintains a grip on the plier handle. The pliers shall be furnished in styles A, B, or C, as specified in 3.10.1.3 through 3.10.1.5, and 6.1. Plier sizes, as specified (see 6.1), shall conform to the dimensions in figure 4 and table II.

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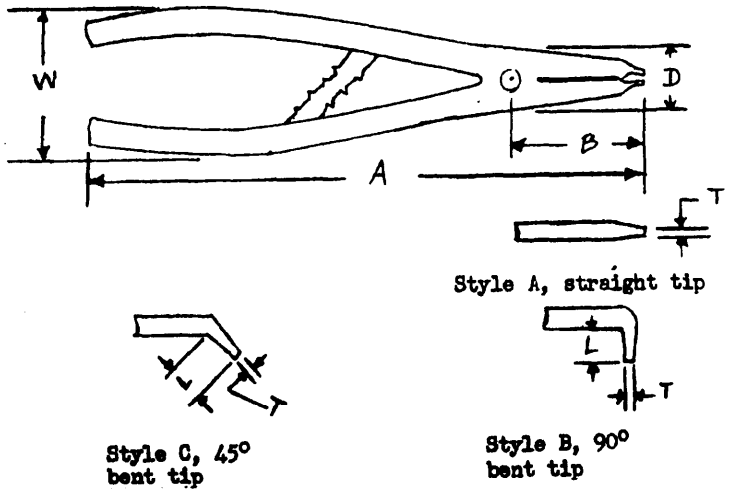
TABLE II. Type I, classes 3 and 4, styles A, B, C, and D pliers, retaining ring, internal, with ratchet, with and without adjustable stop, straight and bent tip

Plier size		Housing diameter of rings		A ±1/2 (See note 2/)	B ±1/4 (See note 3/)	D ±1/4	T	W ±1/2	L ±1/8 Styles B and C	Test ring size
Class 3	Class 4	Basic, bowed or beveled type	Inverted type							
		Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inch	Inches
	1-4-1	1.812 thru 3.000	2.062 thru 3.000	12	2-1/2	1-1/4	0.087 +.003 -.005	4-1/4	7/16	3.000 1/3.000
	1-4-2	3.062 thru 3.562	3.156 thru 3.562	12	2-1/2	1-1/4	0.116 +.003 -.005	4-1/4	7/16	3.562 1/3.562
1-3-3		3.062 thru 6.250	3.156 thru 4.000	16	3-1/2	1-3/4	0.120 ±.005	3-7/8	1/2	6.250 1/3.562
1-3-4		6.500 thru. 7.000	----	19-1/2	5	1-3/4	0.150 ±.005	4	5/8	7.000
1-3-5		7.250 thru 10.000	----	2/34-1/8	3/5	1-3/4	0.150 ±.005	6-1/2	5/8	10.000

1/ Inverted type ring.

2/ Tolerance: ±1 inch on this size plier.

3/ Tolerance: ±1/2 inch on this size plier.



Style A, straight tip

Style C, 45° bent tip

Style B, 90° bent tip

FIGURE 4. Type I, class 3, styles A, B, and C pliers, retaining ring, internal, with ratchet, without adjustable stop, straight and bent tip.

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3.10.4 Type I, class 4, with ratchet and adjustable stop. Type I, class 4 pliers shall conform to the requirements of 3.10.3, except an adjustable stop shall be provided in accordance with 3.10.1.2. The pliers shall be furnished in styles D, B, or C, as specified in 3.10.4.1 through 3.10.4.3, and 6.1. Plier sizes, as specified (see 6.1), shall conform to the dimensions in figure 5 and table II.

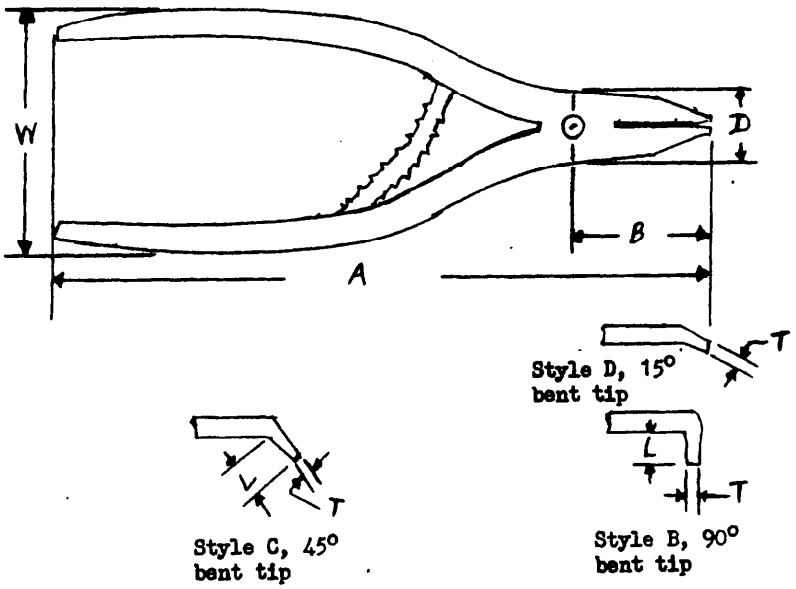


FIGURE 5. Type I, class 4, styles D, B, and C pliers, retaining ring, internal, with double ratchet and adjustable stop, bent tip.

3.10.4.1 Style D, 15° bent tip. The tip end of the jaws of style D pliers shall be bent at a 15° angle to the longitudinal axis of the jaws.

3.10.4.2 Style B, 90° bent tip. The tip end of the jaws of style B pliers shall be bent at a 90° angle to the longitudinal axis of the jaws.

3.10.4.3 Style C, 45° bent tip. The tip end of the jaws of style C pliers shall be bent at a 45° angle to the longitudinal axis of the jaws.

3.11 Type II pliers, retaining ring, external. Type II pliers shall be of steel except as noted. They shall be suitable for expanding external retaining rings similar to those shown in figures 6, 13, or 14, as applicable for the specific class of plier. They shall satisfactorily expand test rings when tested as specified in 4.5.3.



FIGURE 6. External retaining rings.

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3.11.1 Type II, class 1, with fixed jaw stop and spring. Type II, class 1 pliers shall have a fixed outer cover, or stop, to prevent over-expansion of rings. The pliers shall be provided with a spring conforming to 3.10.1.1 which shall be suitable for holding the jaws in a closed position. The pliers shall be furnished in styles A, B, or C, as specified in 3.10.1.3 through 3.10.1.5, and 6.1. Plier sizes, as specified (see 6.1), shall conform to the dimensions shown in figure 7 and table III.

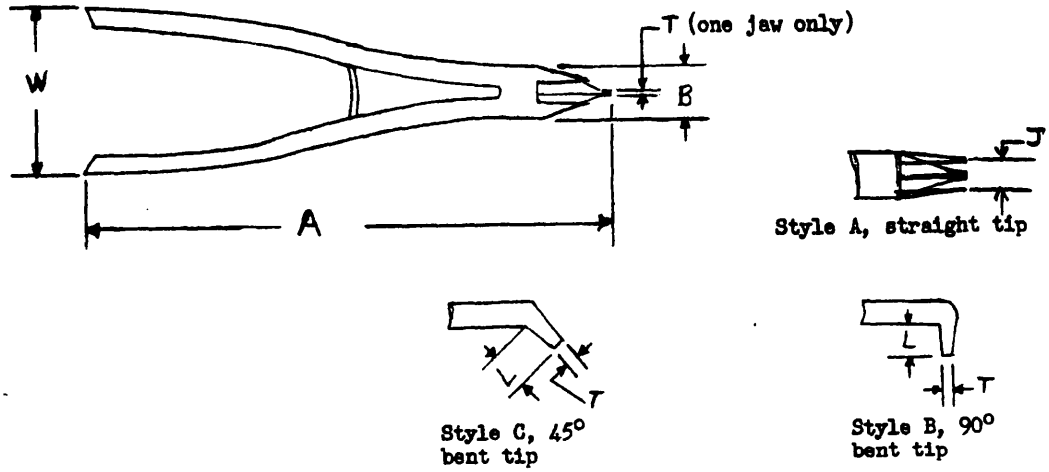


FIGURE 7. Type II, class 1, style A, B, and C pliers, retaining ring, external, with fixed jaw stop and spring, straight and bent tip.

TABLE III. Type II, class 1, style A, B, and C pliers, retaining ring, external, with fixed jaw stop and spring, straight and bent tip

Plier size	Nominal shaft diameter for external rings Basic type Bowed type	A ± 1/8	B ± 1/16	T ±.003	W ± 1/8	J (Style A only)	L ± 1/16 (Style B and C)	Test ring size 1/
	Inch	Inches	Inch	Inch	Inches	Inch	Inch	Inch
2-1-1	0.125	5-5/16	1/2	0.023	1-1/2	0.127 ±.004	7/32	0.125
2-1-2	.156	5-5/16	1/2	.023	1-1/2	.139 ±.006	7/32	.156
2-1-3	.188 thru .236	5-5/16	1/2	.023	1-1/2	.178 ±.008	7/32	.236

1/ Test ring shall be of beryllium copper (4.5.3).

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3.11.2 Type II, class 2, with adjustable stop and spring. Type II, class 2 pliers shall conform to the requirements specified in 3.10.1 through 3.10.1.5, except that they shall be of the external type. Plier sizes, as specified (see 6.1), shall conform to figure 8 and table IV.

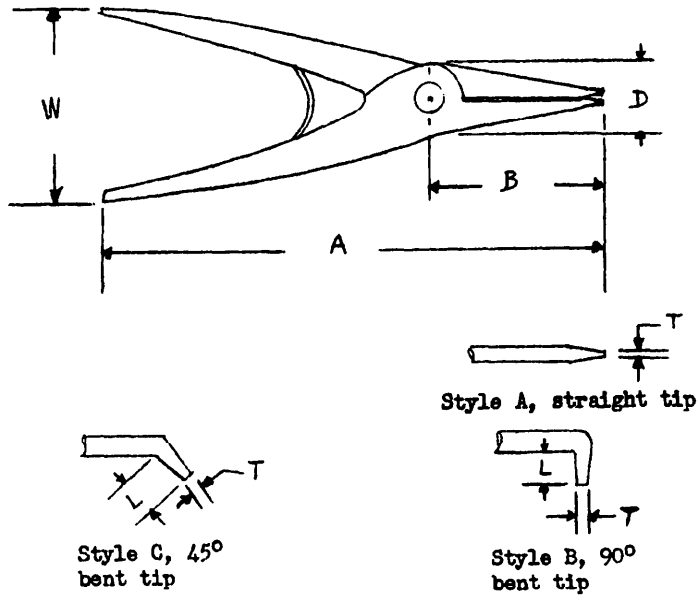


FIGURE 8. Type II, class 2, styles A, B, and C pliers retaining ring, external with adjustable stop and spring, straight and bent tip.

TABLE IV. Type II, classes 2 and 3, styles A, B, and C pliers, retaining ring, external, with and without adjustable stop and spring, straight and bent tip

Plier size		Shaft diameter for external rings		A ±1/2	B ±3/8	D ±3/16	T +.002 -.005	W ±1/2	L (Styles B and C)	Test ring size
Class 2	Class 3	Basic, bowed or beveled type	Inverted type							
		Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inch	Inches
2-2-1	2-3-1	0.250 thru .672	0.500 thru .781	5-7/8	2	1	0.038	2-5/16	9/32 ±1/16	0.669 1/ 0.781
2-2-2	2-3-2	0.688 thru .875	0.812 thru 1.000	5-7/8	2	1	0.047	2-5/16	9/32 ±1/16	0.875 1/ 1.000
2-2-3	2-3-3	0.938 thru 1.438	1.062 thru 2.000	6-13/16	2-3/8	1	0.070	2-3/4	11/32 ±1/16	1.438 1/ 2.000
2-2-4	2-3-4	1.500 thru 3.500	2.125 thru 3.156	9-1/8	3-1/16	1-1/4	0.115	4-1/4	7/16 ±1/8	3.500 1/ 3.156

1/ Inverted type ring.

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3.11.3 Type II, class 3, without adjustable stop and spring. Type II, class 3 pliers shall conform to 3.10.2, except they shall be of the external type. Plier sizes, as specified (see 6.1), shall conform to the dimensions shown in figure 9 and table IV.

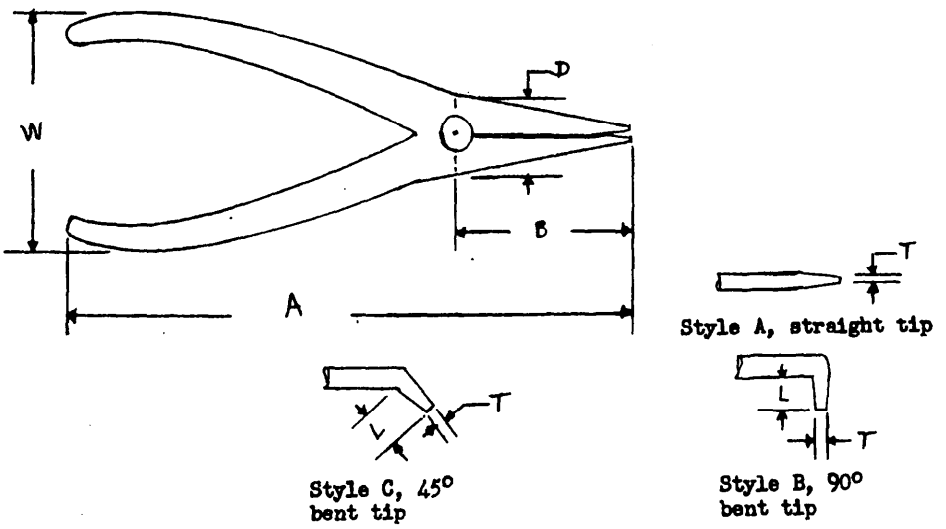


FIGURE 9. Type II, class 3, styles A, B, and C pliers, retaining ring, external, without adjustable stop and spring, straight and bent tip.

3.11.4 Type II, class 4, with ratchet, without adjustable stop. Type II, class 4 pliers shall conform to the requirements of 3.10.3, except they shall be external type. Plier sizes, as specified (see 6.1), shall conform to the dimensions shown in figure 10 and table V.

TABLE V. Type II, classes 4 and 5, styles A, B, C, and D pliers, retaining ring, external, with ratchet, with and without adjustable stop, straight and bent tip

Plier size		Shaft diameter for for external rings			A ±1/2	B ±1/4	D ±1/4	T	W ±1	L ±1/8 Styles B and C	Test ring size
Class 4	Class 5	Basic, bowed or beveled type	Inverted type	Heavy duty type							
		Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inch	Inches
	2-5-1	---	---	1.062 thru 1.750	11-1/2	2-1/2	2-1/4	0.087 +.003 -.005	9	7/16	3/1.750
	2-5-2	1.500 thru 4.375	2.156 thru 3.938	1.938 thru 2.000	11-1/2	2-1/2	2-1/4	0.116 +.003 -.005	9	7/16	1/4.375 3/2.000
2-4-3		3.543 thru 6.500	3.500 thru 3.938	---	15-1/8	3-1/2	2-7/8	0.120 ±.005	11-1/8	5/8	1/6.500 2/3.938
2-4-4		6.750 thru 10.000	---	---	15-1/8	3-1/2	2-7/8	0.150 ±.005	11-1/8	5/8	1/10.000

1/ Basic type ring.
2/ Inverted type ring.
3/ Heavy duty type ring.

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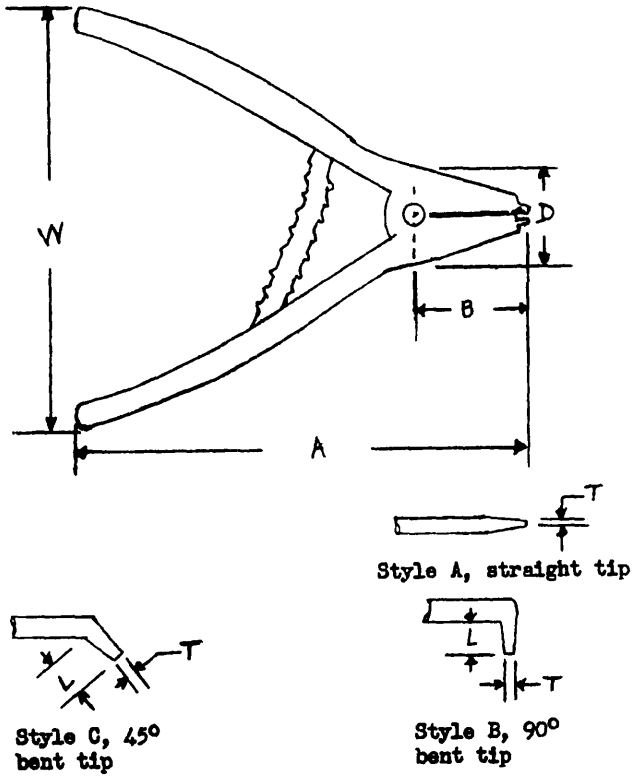


FIGURE 10. Type II, class 4, styles A, B, and C pliers, retaining ring, external, with ratchet, without adjustable stop, straight and bent tip.

3.11.5 Type II, class 5, with ratchet and adjustable stop. Type II, class 5 pliers shall conform to the requirements specified in 3.10.4, except they shall be of the external type. The plier shall be capable of expanding external, heavy duty, retaining rings similar to those shown in figure 14. Plier sizes, as specified (see 6.1), shall conform to the dimensions shown in figure 11 and table V.

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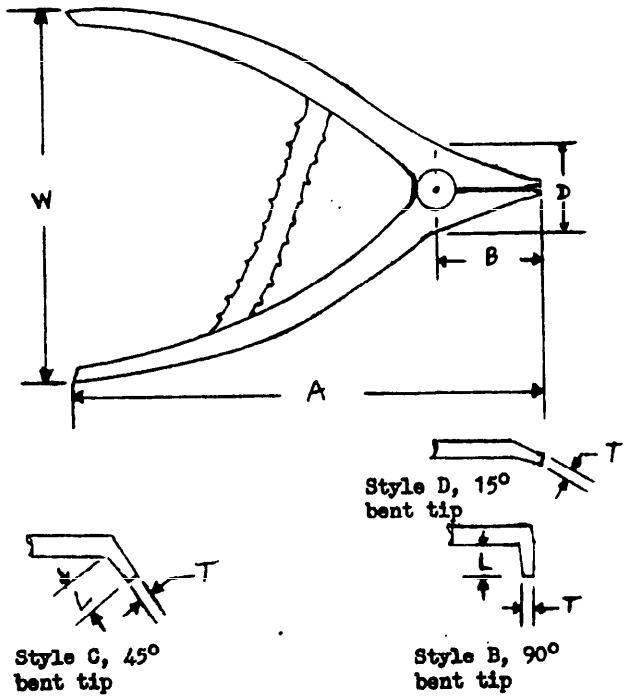


FIGURE 11. Type II, class 5, styles D, B, and C pliers, retaining ring, external, with double ratchet and adjustable stop, bent tip.

3.11.6 Type II, class 6, grip ring pliers, with adjustable stop and spring. Type II, class 6 pliers shall consist of two jaw-handle sections joined by a rivet. An adjustable stop and spring shall be provided. Pliers shall be furnished in styles E, B, or C, as specified in 3.11.6.1 through 3.11.6.3, and 6.1. Pliers shall be suitable for expanding external grip rings and external heavy duty type rings similar to those shown in figures 13 and 14. Plier sizes, as specified (see 6.1), shall conform to the dimensions in figure 12 and table VI.

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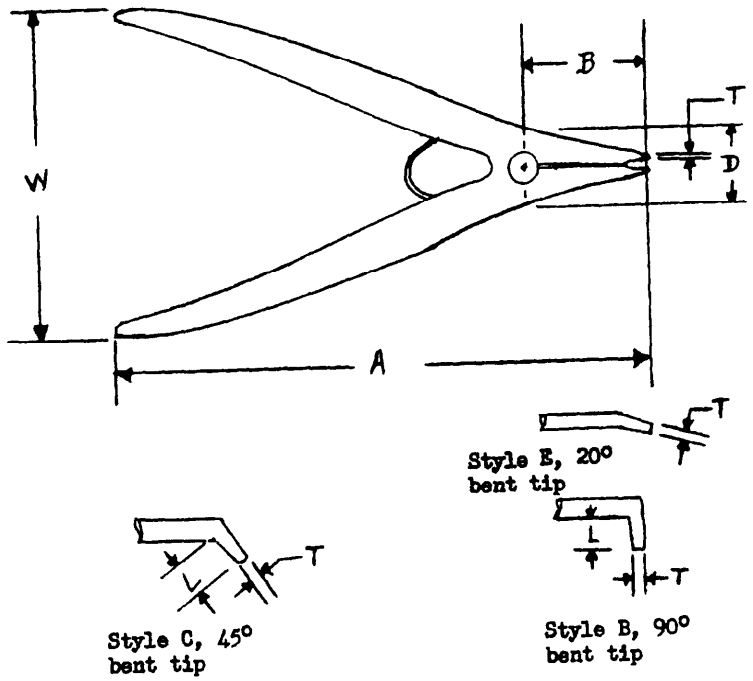


FIGURE 12. Type II, class 6, styles E, B, and C pliers, retaining ring, external grip, with adjustable stop and spring, with bent tips.

TABLE VI. Type II, class 6, styles E, B, and C pliers, retaining ring, external grip, with adjustable stop and spring, with bent tips

Plier size	Nominal shaft diameter for external rings		A ± 1/2	B ± 1/4	D ± 3/16	T +.002 -.005	W ± 1/2	L (Styles B and C)	Test ring size
	Grip type	Heavy duty type							
	Inch	Inch	Inches	Inches	Inch	Inch	Inches	Inches	Inch
2-6-1	0.094	- - -	5	1-1/16	5/8	0.034	2-7/8	5/16 ± 1/16	0.094
2-6-2	0.125 thru .156	0.394 thru .473	5-1/16	1-1/16	5/8	0.040	2-7/8	5/16 ± 1/16	0.156 1/ .473
2-6-3	0.187 thru .250	0.500 thru .669	5-1/16	1-1/16	5/8	0.047	2-7/8	5/16 ± 1/16	0.250 1/ .669
2-6-4	0.313 thru .750	0.750 thru .984	8-7/8	1-5/8	1	0.070	3-7/8	3/8 ± 1/8	0.750 1/ .984

1/ Heavy duty type ring.

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FIGURE 13. External grip ring.

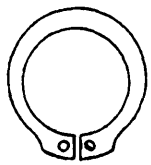


FIGURE 14. External heavy duty type ring.

3.11.6.1 Style E, 20° bent tip. The tip end of the jaws of style E pliers shall be bent at a 20° angle to the longitudinal axis of the jaws.

3.11.6.2 Style B, 90° bent tip. The tip end of the jaws of style B pliers shall be bent at a 90° angle to the longitudinal axis of the jaws.

3.11.6.3 Style C, 45° bent tip. The tip end of the jaws of style C pliers shall be bent at a 45° angle to the longitudinal axis of the jaws.

3.12 Type III, pliers retaining ring, external (automotive). The pliers shall be provided with a spring for holding the jaws in a closed position. The spring shall be in accordance with 3.10.1.1. The pliers shall have a minimum Rockwell C hardness of 30 when tested as specified in 4.5.5.

3.12.1 Class 1, straight jaw. Type III, class 1 pliers shall have long tapered jaws, used primarily in removing and synchronizing snap rings in close places. The outside of the jaws at the tip end shall be serrated or knurled. The pliers shall be similar in design, and conform to the dimensions shown on figure 15.

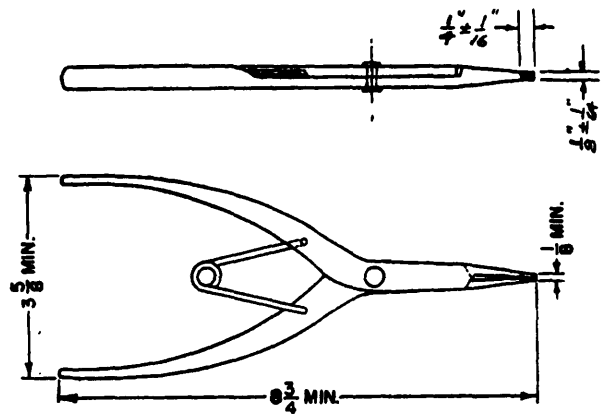


FIGURE 15. Type III, class 1 pliers, retaining ring, external (automotive), straight jaw.

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3.12.2 Class 2, brake key. Type III, class 2 pliers shall be capable of expanding (horseshoe) "C" washers and lock rings, on automotive brakes, transmissions, differentials, and pulleys. The tip end of the jaws shall be at right angles to the longitudinal axis of the jaws, and the outside serrated or knurled to prevent slipping. The pliers shall conform to the dimensions shown in figure 16 and table VII.

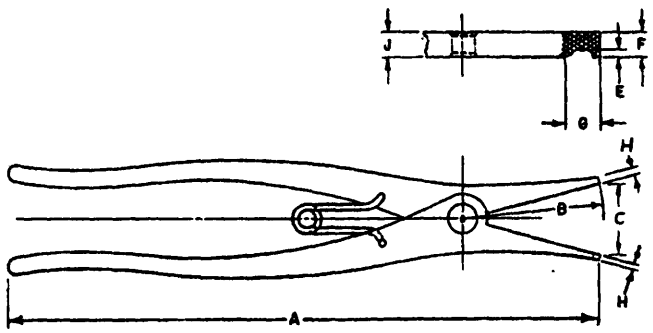


FIGURE 16. Type III, class 2 pliers, retaining ring, external (automotive), brake key.

TABLE VII. Type III, class 2 pliers, retaining ring, external (automotive) brake key

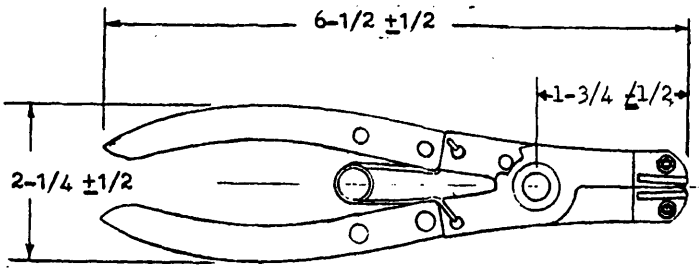
Plier size	A Min.	B Max.	1/ C Min.	C Max.	E Max.	F Max.	G Min.	H ± 0.010	J Max.
	Inches	Inches	Inch	Inch	Inch	Inch	Inch	Inch	Inch
3-2-1	7	1-7/8	25/32	1-7/16	1/8	3/8	7/16	0.050	3/8

1/ The "C" dimension shall be measured when the handles of the assembled tool are in a fully-closed position.

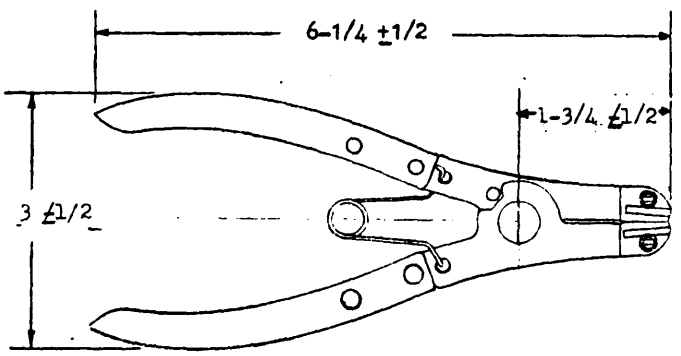
3.13 Type VII plier sets with interchangeable points. Type VII plier sets shall consist of two pliers. One for compressing rings similar to figure 1 and the other for expanding rings similar to figure 6. One set of interchangeable points shall be included. The pliers shall satisfactorily contract or expand test rings when tested as specified in 4.5.3. Each set shall be furnished in a plastic bag or box, which will include the points packaged in a plastic box.

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3.13.1 Class 1, with spring. Type VII, class 1 pliers shall be made of two jaw-handle sections, joined by a rivet, and shall be provided with a spring. The spring shall conform to 3.10.1.1. The pliers shall conform to the dimensions shown in figure 17. The plier jaws shall have a minimum inside spread of 5/8 inch when the jaws are fully opened.



Internal snap ring plier



External snap ring plier (in closed position).

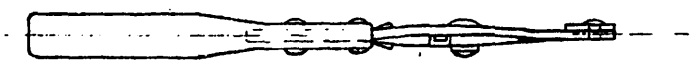


FIGURE 17. Type VII, class 1 plier set, with spring.

3.13.2 Jaws. The jaw ends shall be capable of receiving and securely holding interchangeable points in correct alignment.

3.13.3 Interchangeable points. Interchangeable points shall be furnished in sets of two, one for each jaw. The points shall be steel, hex stock, heat treated to Rockwell C hardness of 45 to 50 when tested as specified in 4.5.5. The point tips shall be properly machined to provide for gripping and securely holding the retaining rings. Tip ends shall be round in cross section. Each set of points shall be interchangeable for use with both pliers in the set. A set of points, numbers 1 through 9, shall be furnished with each set of class 1 pliers, and shall conform to the dimensions in table VIII for the tip size required and be similar to figure 18.

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TABLE VIII. Interchangeable points for type VII, class 1 pliers.

Point number	A Tip angle ∠ 5°	1/B Tip diameter	2/C Tip length	Nominal housing diameter for internal rings		Nominal shaft diameter for external rings	
				Basic type	Inverted type	Basic type	Inverted type
	Degrees	Inches	Inches	Inches	Inches	Inches	Inches
1	15	0.039	0.059	0.375	0.750	0.250	0.500
2	45	∠0.000	∠0.010	through	through	through	through
3	90	-0.007		0.438	1.000	0.469	0.781
4	15	0.047	0.074	0.453	1.062	0.500	0.812
5	45	∠0.000	∠0.010	through	through	through	through
6	90	-0.007	-0.014	0.562	1.375	0.875	1.000
7	15	0.074	0.086	1.062	1.438	0.938	1.062
8	45	∠0.000	∠0.022	through	through	through	through
9	90	-0.017	-0.010	1.750	2.000	1.438	2.000

1/ 0.039-inch dia. point identified by plain shank, or color differentation.
.047-inch dia. point identified by one notch on shank, or color differentation.
.074-inch dia. point identified by two notches on shank, or color differentation.

2/ Points with 15° and 45° angle excludes radius dimension.
Points with 90° angle includes radius dimension.

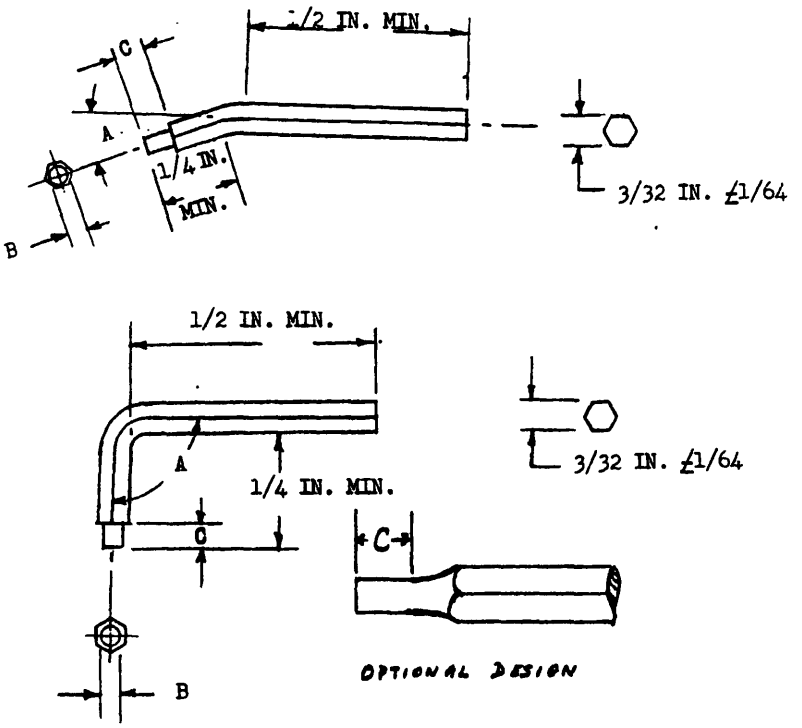


FIGURE 18. Interchangeable points for type VII, class 1 and 2 pliers.

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3.13.4 Class 2, with ratchet. Type VII, class 2 pliers, shall be made of two jaw-handle sections, joined by a rivet, and shall be provided with a ratchet. The ratchet shall be of steel, and designed in such a manner that the jaws may be locked and released in any position within the spreading or expanding capacity of the pliers. The locking arrangement of the ratchet shall be capable of being released with the hand maintaining a grip on the handle. The pliers shall conform to the dimensions shown in figure 19. The plier jaws shall have a minimum inside spread of 1 inch when the jaws are fully opened.

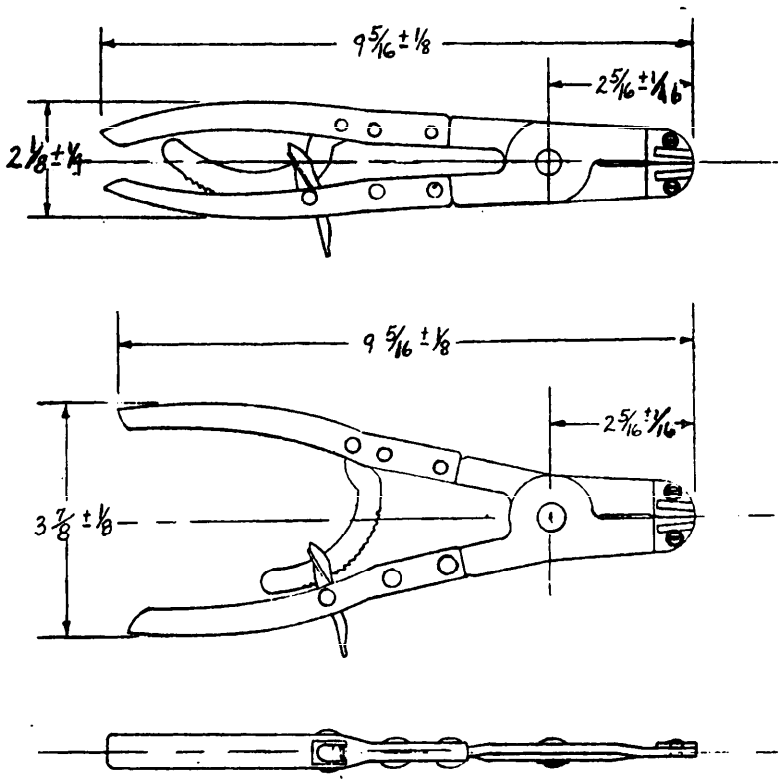


FIGURE 19. Type VII, class 2 plier set with ratchet.

3.13.4.1 Interchangeable points. Interchangeable points, numbers 10 through 12, shall conform to the dimensions in table IX, and be similar to figure 18. Points shall be heat treated to a Rockwell C hardness of 50 to 55 when tested as specified in 4.5.5. A set of points, numbers 10, 11, and 12 shall be furnished with each set of class 2 pliers.

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TABLE IX. Interchangeable points for type VII, class 2 pliers

Point number	1/ A Tip dia. at point ± 0.000 - .005	B Tip angle ± 5°	C Tip length ± 0.010	2/ D Length straight portion ± 1/32	2/ E Length straight portion ± 1/32	F Size of hex stock ± 1/64	Nominal housing diameter for internal rings		Nominal shaft diameter for external rings	
							Basic type	Inverted type	Basic type	Inverted type
	Inches	Degrees	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
10	0.108	15	0.129	0.350	63/64	5/32	2.440 through 3.125	2.625 through 3.000	1.500 through 3.500	2.125 through 3.346
11	.108	45	.129	.470	61/64	5/32				
12	.123	45	1.290	.470	61/64	5/32	3.156 through 4.000	3.156 through 4.000	1.562 through 3.750	3.5000 through 3.750

1/ 0.108-inch dia. point identified by plain shank, or color differentation.
.123-inch dia. point identified by one notch on shank, or color differentation.

2/ Points with 15° and 45° angle excludes radius dimension.
Points with 90° angle includes radius dimension.

3.14 Type VIII pliers, retaining ring, internal and external (universal). Type VIII pliers shall be of steel suitable for compressing internal retaining rings similar to those shown in figure 1, and expanding external rings similar to those shown in figures 6 and 13. They shall satisfactorily compress or expand test rings when tested as specified in 4.5.2.

3.14.1 Type VIII, class 1, without adjustable stop and spring. Type VIII, class 1 pliers shall consist of two jaw-handle sections which may be assembled in the field to expand external retaining rings (external setting) or compress internal retaining rings (internal setting). Plier sizes, as specified (see 6.1), shall conform to the dimensions in figure 20 and table X. The pliers shall be furnished in styles A, B, or C, as specified in 3.10.1.3 through 3.10.1.5, and 6.1.

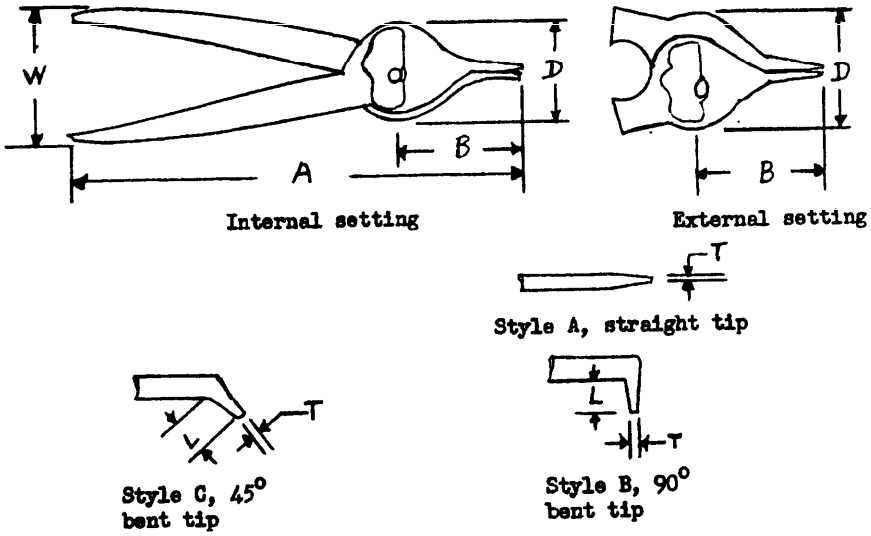


FIGURE 20. Type VIII, class 1, styles A, B, and C pliers, retaining ring, internal and external (universal), straight and bent tip.

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TABLE X. Type VIII, class 1, styles A, B, and C pliers, retaining ring, internal and external (universal), straight and bent tip

Plier size	Plier style	Housing diameter for internal rings		Shaft diameter for external rings			A ± 1/2	B ± 1/4	D ± 3/16	T +.002 -.005	W ± 1/2	L ± 1/16 style B and C only	Test ring size
		Basic, bowed, beveled type	Inverted type	Basic, bowed, beveled type	Inverted type	Grip type							
		Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inches	Inch	Inches
8-1 8-2 8-3	A B C	0.375 thru 1.023	0.750 thru 1.375	0.250 thru 0.875	0.500 thru 1.000	0.125 thru 0.250	5-5/8	1-5/8	1-3/16 Internal setting 1-3/8 External setting	0.038	1-3/4 Internal setting 2-9/16 External setting	9/32	1/ 1.023 2/ 1.375 3/ .875 4/ 1.000
8-4 8-5 8-6	A B C	1.062 thru 1.750	1.438 thru 2.000	0.938 thru 1.438	1.062 thru 2.000	0.437 thru 0.750	7-7/8	1-7/8	1-7/16 Internal setting 1-11/16 External setting	0.070	3-5/16 Internal setting 3-15/16 External setting	11/32	1/ 1.750 2/ 2.000 3/ 1.438 4/ 2.000

1/ Basic internal type ring.
2/ Inverted internal type ring.
3/ Basic external type ring.
4/ Inverted external type ring.

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3.14.1.1 Latching means. Type VIII, class 1 pliers shall be provided with latching means which, when engaged, prevent the two jaw-handle sections from becoming disassembled when the plier is in use. When the latching means is disengaged, disassembly of the plier for changing the setting shall be possible.

3.14.1.2 Construction. Type VIII, class 1 pliers shall have all parts captive with either of the two jaw-handle sections. It shall not be necessary to detach any parts from either of the two jaw-handle sections when they are disassembled and reassembled for the purpose of changing the plier setting.

3.15 Workmanship. The pliers shall conform to the quality and grade of product covered by this specification. The pliers shall be free from rust, laps, cracks, blisters, and other defects which may impair their serviceability, durability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling for lot acceptance. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table XI.

TABLE XI. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	One complete plier, or one plier set	II	2.5 for majors 6.5 for total	Defects per hundred units	4.3.1
Dimensional examination	One complete plier, or one plier set	II	2.5	Percent defective	4.3.2
Testing (Including dimensional requirements with 3 or more decimal places)	One complete plier, or one plier set	S-2	1.5	Percent defective	4.5
Examination of preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.6

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4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table XII.

TABLE XII. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Coating not type specified.	X	
	Surface not usually ground or buffed, does not conform to best manufacturing practices.		X
	Not free from pits, nodules, laps, blisters.		X
	Rust, cracks, or scratches that may impair serviceability or durability.	X	
Design	Any characteristic not in accordance with the specified requirements.	X	
Rivets	Missing, cracked, or improperly peened (if tubular rivets are used, they shall be limited to not more than two minor cracks in each rivet head).	X	
Welding	Missing, fractured, cracked, or not fused (type I).	X	
	Not smooth, gas pockets, pits, holes undercut, or burned through.		X
Construction and workmanship	Component missing or not of specified type or size.	X	
	Component in wrong location, loose, or not in proper alignment.		X
	Not connected or joined as specified or poorly accomplished.	X	
	Fractured, split, binding spring, malformed, or otherwise impaired.	X	
	Sharp burr or sliver on surface of handles.	X	
	Adjustable component will not function as specified.	X	
Marking	Missing, incomplete, not legible.		X

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.5.1 through 4.5.5, as applicable.

4.5 Test procedures.

4.5.1 Types I and VII pliers. The test rings, used for contraction tests, shall be in accordance with MIL-R-21248, type I, classes 1 and 3. One contraction of three test rings, of each test ring size, shall be sufficient to determine the acceptability of the pliers. The plier tips shall be adequately inserted in the test ring plier holes. The ring shall then be contracted by pressure on the plier handles until the ring lugs contact each other. With the ring lugs contacting each other, the ring shall not become dislodged from the plier tips if the plier operator removes his hand from the ring. The test ring shall then be released. One ring of the smallest, and one ring of the largest size in the applicable size range of the pliers, shall be held loosely between the thumb and forefinger, while the plier tips are inserted in the ring holes, to insure that there was no plier tip deformation during the ring contraction test, thus making tip insertion in the ring holes difficult. The plier shall also be examined for any evidence of cracks or breakage.

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4.5.2 Type VIII pliers. The test procedure for type VIII pliers shall consist of contraction tests in accordance with 4.5.1 followed by expansion tests in accordance with 4.5.3. Test rings shall be in accordance with MIL-R-21248, type I, classes 1, 2, 3, and 4. The plier shall first be assembled in the internal position for the contraction tests. This test shall be followed by disassembly and reassembly of the plier in the external position. Expansion tests shall then be performed, followed by a check for tip deformation in accordance with 4.5.3. There shall be no deformation during these tests which would prevent assembly or disassembly of the plier for purposes of changing the setting.

4.5.3 Type II and VII pliers. The test rings used for expansion tests shall be in accordance with MIL-R-21248, type I, classes 2, 4, 7, and 10 (see note 1 of table III). One expansion of three test rings of each test ring size shall be sufficient to determine the acceptability of the pliers. The plier tips shall be inserted adequately in the test plier holes. The ring shall then be expanded by pressure on the plier handles until the internal diameter of the ring is one percent greater than the diameter of the shaft over which the ring is designed to pass during installation. One ring of the smallest and one ring of the largest size, in the applicable size range of the pliers, shall be held loosely between the thumb and forefinger, while the plier tips are inserted in the ring holes, to insure that there was no plier tip deformation during the ring expansion test, thus making tip insertion in the ring holes difficult. The pliers shall also be examined for any evidence of cracks or breakage.

4.5.4 Type III pliers. Type III pliers shall be tested for load in the following manner. A platform scale similar to figure 21 may be used for making load tests. With the handles fully closed and clamped, the jaws will be at the maximum open position. The handles shall be placed in a normal horizontal position resting on a support. One tip end shall rest on a knife edge support and a test load of 30 pounds shall be applied to the other tip. After testing, the pliers shall not show any evidence of permanent set, cracks, or breakage.

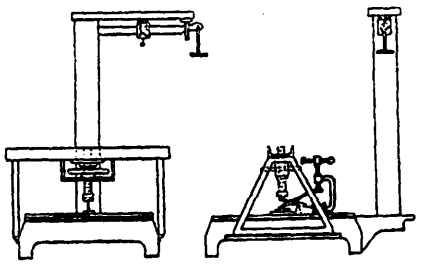


FIGURE 21. Apparatus for applying test load to tips of type III pliers.

4.5.5 Hardness test. The Rockwell hardness tests shall be conducted in accordance with method 243.1 of Fed. Std. No. 151. Type III pliers shall be tested on the flat surface of the jaw. Interchangeable points shall be tested anywhere on the hex portion.

4.6 Inspection of preparation for delivery. Inspection shall be made to determine compliance with the requirements of PPP-P-40.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style of plier required (see 1.2).
(c) Size of plier required (see applicable table).
(d) Specify level of preservation, packaging, and packing (see 5.1).

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